

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data to Delta Tables

Title: Load Customer and Order Data to Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

CSV files are available at the specified locations:

``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and
``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``

- The Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and load it into a Delta table named "customer".
2. Read order CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`` and load it into a Delta table named "order".
3. Add a new column "TotalAmount" to the "order" table, which is the product of "PricePerUnit" and "Qty" columns.

FR-DLT-002: Clean and Process Customer and Order Data

Title: Clean and Process Customer and Order Data

Description: The system must remove null and duplicate records from the customer and order Delta tables.

Preconditions:

- Customer and order Delta tables exist and contain data

Main Flow / Functional Steps:

1. Remove null records from the "customer" and "order" Delta tables.
2. Remove duplicate records from the "customer" and "order" Delta tables.

FR-DLT-003: Create and Load ordersummary Table

Title: Create and Load ordersummary Table

Description: The system must create an "ordersummary" table if it does not exist and load it with joined customer and order data using SCD Type 2.

Preconditions:

- Customer and order Delta tables exist and contain clean data
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "ordersummary" table if it does not exist in the specified catalog and schema (`gen_ai_poc_databrickscoe.sdmc_wizard`).
2. Join the "customer" and "order" Delta tables on the "CustId" column.
3. Load the joined data into the "ordersummary" table using SCD Type 2.

FR-DLT-004: Update ordersummary Table on Customer Changes

Title: Update ordersummary Table on Customer Changes

Description: The system must update the "ordersummary" table whenever there are changes to the customer data.

Preconditions:

- "ordersummary" table exists and contains data
- Customer data has changed

Main Flow / Functional Steps:

1. Identify changes to the customer data.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update the "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-005: Create and Load customeraggregatespend Table

Title: Create and Load customeraggregatespend Table

Description: The system must create a "customeraggregatespend" table if it does not exist and load it with aggregated data from the "ordersummary" table.

Preconditions:

- "ordersummary" table exists and contains data

- "customeraggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create the "customeraggregatespend" table if it does not exist in the specified catalog and schema (`gen\_ai\_poc\_databrickscoe.sdlc\_wizard`).
2. Aggregate the "TotalAmount" column from the "ordersummary" table by "Name" and "Date" columns.
3. Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-006: Implement Delta Live Tables

Title: Implement Delta Live Tables

Description: The system must implement the above requirements using Delta Live Tables.

Preconditions:

- All previous requirements have been implemented

Main Flow / Functional Steps:

1. Implement the data pipelines using Delta Live Tables to load and process customer and order data.
2. Use Delta Live Tables to create and maintain the "ordersummary" and "customeraggregatespend" tables.